Technical Report

pyscdblfinder: a Python implementation of scDblFinder for doublet detection in single-cell data

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Summary

We present pyscdblfinder, a Python implementation of scDblFinder for doublet detection in single-cell RNA-seq and ATAC-seq data, covering artificial-doublet generation, co-expression and nearest-neighbour features, the iterative gradient-boosted classifier, thresholding against the expected doublet rate, feature aggregation for chromatin data and AMULET. pyscdblfinder follows scDblFinder's code paths step by step and is versioned against a specific scDblFinder release; the steps that use no random numbers return the same values as R. On the 16 real datasets of an independent doublet-detection benchmark (168,239 cells, 3–37% doublets annotated experimentally), pyscdblfinder detects annotated doublets as accurately as scDblFinder 1.16.0 (mean area under the precision–recall curve 0.560 versus 0.562), and its scores agree with scDblFinder's as closely as two runs of scDblFinder with different seeds (median Spearman correlation 0.874 in both cases), while running 1.1–2.5 times faster on one thread. pyscdblfinder is the default doublet detector of OmicVerse quality control and is released under GPL-3.0 at `https://github.com/omicverse/py-scDblFinder` and on PyPI.

Keywords scDblFinder, doublet detection, single-cell RNA-seq, quality control, xgboost, Python, reproducibility

Introduction

Droplet-based single-cell sequencing captures two or more cells in the same droplet at a rate that grows with the number of cells loaded, and these doublets are a routine quality-control concern[1]. Doublets of different cell types look like intermediate or novel cell states and can distort clustering, trajectories and differential expression, so computational doublet detection is a standard step of single-cell analysis[1,2]. scDblFinder[3] mixes real cells with artificial doublets made from random pairs of cells and describes every cell by the composition of its neighbourhood, a co-expression score[4] and library-size features. A gradient-boosted tree classifier[5] then learns to tell the two apart, refining its training set over several iterations. On the real datasets of the benchmark of Xi and Li[2] it was among the most accurate methods in its authors' comparison[3].

Single-cell analysis is increasingly done in Python. Workflows are built on scanpy and the scverse ecosystem[6,7], and OmicVerse[8] brings bulk, single-cell and spatial analyses together in one Python framework. Python offers doublet detectors of its own, such as Scrublet[9], but scDblFinder is available only in R.

Python users currently reach scDblFinder through rpy2[10]. This requires a parallel R and Bioconductor installation, converts the count matrix between the two languages and exposes scDblFinder's options only through the binding layer. A native implementation removes all three costs, as PyDESeq2[11] has shown for DESeq2[12] and pyedger[13] for edgeR[14]. We set the same target for pyscdblfinder: a doublet analysis in Python should detect doublets as accurately as scDblFinder and agree with it as closely as scDblFinder agrees with itself.

Here we present pyscdblfinder, a Python implementation of scDblFinder that is versioned against a specific scDblFinder release. We describe its implementation and compare it with scDblFinder 1.16.0 on the 16 real datasets of the Xi and Li benchmark[2].

Implementation

Method

scDblFinder first generates artificial doublets by summing the counts of random pairs of cells, by default 80% as many as there are real cells (at least 1,500 and at most 25,000). Real and artificial cells are then reduced to the most highly expressed genes, log-normalised by library size[15] and projected on their first 20 principal components. For every cell, scDblFinder records the proportion of artificial doublets among its nearest neighbours for a grid of neighbourhood sizes, a distance-weighted version of this proportion and the distances to the nearest real and artificial cells. It adds the cxds co-expression score[4], the library size and the number of detected genes. A gradient-boosted tree classifier (xgboost[5]) is trained on these features to separate real from artificial cells, with the number of boosting rounds chosen by cross-validation. Training is repeated three times; after each round the real cells already

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scored as likely doublets are removed from the training set. Finally, a score threshold is chosen that balances the misclassification of real and artificial cells against the deviation of the number of called doublets from the expected doublet rate, which by default grows by 1% per thousand cells[3].

Features and code structure

pyscdblfinder 0.3.2 implements the single-sample workflow of scDblFinder 1.16 (Table 1): artificial-doublet generation with its library-size adjustments, the cxds score, the kNN features, the iterative classifier and the thresholding, as well as optional user-supplied clusters. For chromatin data it implements scDblFinder's feature aggregation with TF-IDF normalisation, AMULET[16] on fragment files and from counts, the clamulet classifier and the combination of scDblFinder and AMULET by Fisher's method. `sc_dbl_finder` takes a features × cells matrix like scDblFinder; the `ScDblFinder` class runs on an AnnData object[17] and writes `scDblFinder_score` and `scDblFinder_class` to its cell table. pyscdblfinder builds on NumPy[18], SciPy[19], pandas[20], scikit-learn[21] and the xgboost Python package[5], and runs on one thread by default, as scDblFinder does.

Numerical parity with scDblFinder

pyscdblfinder ports scDblFinder's code paths instead of re-deriving the method. These include the choice of genes by mean count with R's ordering of ties, scuttle's log-normalisation[15], and scDblFinder's rules for pairing, halving and resampling artificial doublets. They also include its grid of neighbourhood sizes, the rules that exclude cells from training, the cross-validated choice of boosting rounds with exact tree construction, and the thresholding objective, which is minimised with Brent's method as in R's `optimize`. Given the same scores, pyscdblfinder returns scDblFinder's threshold and doublet calls; the steps without random numbers are tested against stored output of scDblFinder 1.16.0, and the full test suite has 33 tests. Artificial doublets and xgboost's training draw random numbers, which R and Python generate differently, so scores from the two languages, like scores from two R seeds, are not identical cell by cell. pyscdblfinder versions follow scDblFinder releases (0.3.x tracks scDblFinder 1.16.0).

Use in OmicVerse

pyscdblfinder is the default doublet detector of OmicVerse[8] quality control: `ov.pp.qc(adata, doublets_method='scdblfinder')` runs it on the raw counts, once per sample when `batch_key` is given, and writes the doublet scores and calls to `adata.obs`.

Comparison with scDblFinder on real data

Data and design

We used the 16 real datasets of the Xi and Li doublet-detection benchmark[2], in which doublets were annotated experimentally by demultiplexing genotypes with demuxlet[22] (6 datasets), cell hashing[23] (3), MULTI-seq[24] (5) or mixing human and mouse cells (2). The datasets have 500 to 26,426 cells (168,239 in total), of which 3 to 37% are annotated doublets. Both packages ran with default settings on the counts as provided, scDblFinder 1.16.0 from R and pyscdblfinder from Python, each with seeds 1 and 2. We compared four criteria (Supplementary methods). The first is the detection of annotated doublets, measured by the area under the precision–recall curve (AUPRC[25]) and the ROC curve (AUROC). The second is the agreement of the doublet scores (Spearman correlation) and calls (Jaccard index) between Python and R, set against the agreement between the two R seeds. The last two are the fraction of cells called doublets and run time.

Doublet detection and agreement with scDblFinder

pyscdblfinder detected the annotated doublets as accurately as scDblFinder (Fig. 1a). Averaged over datasets and seeds, the AUPRC was 0.560 for pyscdblfinder and 0.562 for scDblFinder, and the AUROC 0.794 and 0.796. Per dataset the AUPRC of the two packages differed by at most 0.040, while the AUPRC of scDblFinder itself varied by up to 0.042 between its two seeds.

Python and R agreed with each other as closely as two R runs with different seeds (Fig. 1b, c). The median Spearman correlation of the doublet scores was 0.874 between Python and R and 0.874 between R seeds, and the ratio of the two was 1.00 at the median and 0.92 at the lowest. The cells called doublets overlapped with a median Jaccard index of 0.768 between Python and R and 0.770 between R seeds. The lowest agreement was on J293t-dm, the smallest dataset (500 cells, 16 doublets called by R), where the Jaccard index was 0.56 between Python and R and 0.72 between R seeds.

On pbmc-2ctrl-dm, a dataset of 13,913 peripheral blood cells with 1,598 doublets annotated by demuxlet, the two packages mark the same cells in the embedding (Fig. 1f–i). High scores from both packages fall on the annotated doublets scattered through the cell-type clusters (Spearman correlation of the scores 0.89), and 1,878 cells were called doublets by both packages, 93 by scDblFinder only and 69 by pyscdblfinder only.

Called doublets and speed

Both packages called 3 to 16% of cells doublets, and per dataset their called fractions differed by at most 0.8 percentage points (median 0.2; Fig. 1e). Where the annotated fraction was far above the expected doublet rate, as in HMEC-rep-MULTI and mkidney-ch, both called fewer doublets than were annotated.

pyscdblfinder ran 1.1 to 2.5 times faster than scDblFinder (median 1.4), both on one thread (Fig. 1d). For HMEC-orig-MULTI, the largest dataset (26,426 cells), scDblFinder took 241 s and pyscdblfinder 190 s.

Conclusion and future perspectives

pyscdblfinder brings scDblFinder's doublet detection to Python. On 16 real datasets with experimentally annotated doublets it is as accurate as scDblFinder 1.16.0, agrees with it as closely as scDblFinder agrees with itself across seeds, calls the same fraction of doublets and runs faster on one thread. Because pyscdblfinder works on AnnData objects and is the default doublet detector of OmicVerse quality control, doublet removal can run in the same Python session as the rest of a single-cell workflow.

Future work includes the parts of scDblFinder beyond the single-sample workflow: the multi-sample modes, the use of known doublets, cluster-correlation features, and the functions `recoverDoublets`, `findDoubletClusters` and `computeDoubletDensity`. pyscdblfinder will continue to follow scDblFinder releases, with each version pinned to the release it reproduces.

Table 1. scDblFinder steps and their pyscdblfinder counterparts.

scDblFinder	pyscdblfinder	Agreement with scDblFinder 1.16.0
gene selection	sc_dbl_finder	same genes, same order
library-size features	sc_dbl_finder	identical
.defaultKnnKs	r_default_knn_ks	same neighbourhood sizes
doubletThresholding	doublet_thresholding	threshold, relative difference below 10^{-9}
getArtificialDoublets	get_artificial_doublets	same pairing rules; random pairs
scDblFinder (scores)	sc_dbl_finder	Spearman 0.874 with R; R seed to seed 0.874
scDblFinder (calls)	sc_dbl_finder	Jaccard 0.768 with R; R seed to seed 0.770

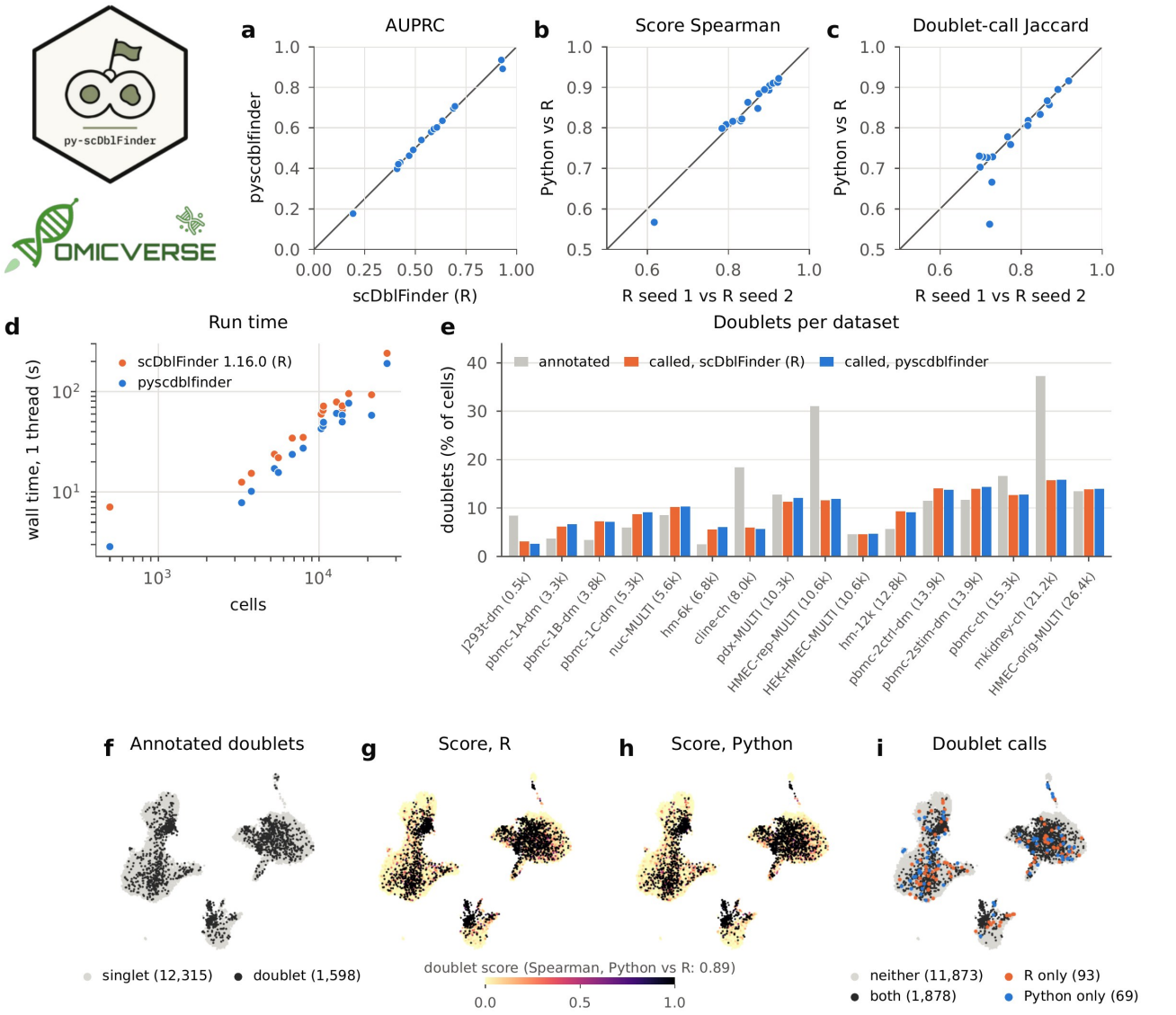


Figure 1. pyscdblfinder versus scDblFinder 1.16.0 on the 16 real datasets of the Xi and Li benchmark.

(a) Area under the precision–recall curve for the annotated doublets; each point is one dataset, the mean of two seeds, with bars from the lower to the higher seed. **(b)** Spearman correlation of the doublet scores of all cells, between Python and R (y; mean of the four seed pairs) against between R seeds 1 and 2 (x). **(c)** As (b), for the Jaccard index of the cells called doublets. **(d)** Wall time on one thread against the number of cells (mean of two seeds). **(e)** Annotated doublets and cells called doublets by each package, as a percentage of all cells; datasets are ordered by size, with the number of cells in parentheses. **(f–i)** UMAP[26] of the 13,913 cells of pbmc-2ctrl-dm, computed with scanpy from the counts: **(f)** doublets annotated by demuxlet; **(g, h)** doublet scores of scDblFinder and pyscdblfinder (seed 1 each), cells with higher scores drawn on top; **(i)** cells called doublets by both packages, by one of them, or by neither.

Data and code availability

pyscdblfinder is released under GPL-3.0 on PyPI (`pip install pyscdblfinder`) and at https://github.com/omicverse/py-scDblFinder; this report describes version 0.3.2. The benchmark was run with release 0.3.1, whose `sc_dbl_finder` gives results identical to 0.3.2. The benchmark (`report/scripts` in the repository) reproduces every number and figure in this report. The Xi and Li real datasets were obtained from Zenodo (https://doi.org/10.5281/zenodo.4562782), accessed on 9 November 2026. Software: pyscdblfinder (Python 3.10, NumPy 2.2.6, SciPy 1.15.2, scikit-learn 1.7.2, xgboost 3.0.5; scanpy 1.11.5 for Fig. 1f–i); scDblFinder 1.16.0 with xgboost 1.7.6.1 on R 4.3.3 (bioconda). Hardware: one node of the Stanford Sherlock cluster (Intel Xeon Platinum 8462Y+, 20 cores allocated).

Declaration of generative AI and AI-assisted technologies

The authors used Claude Opus 5.5, GPT-6 Astra, and OmicOS to proofread grammar and syntax, improve readability. Any AI-assisted edits were reviewed by the authors, who take full responsibility for the content of this publication.

Conflicts of interest

Z.Z. and X.M. are affiliated with OmicVerse, Inc.

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Supplementary information

Supplementary methods

Data. The 16 datasets of `real_datasets.zip` (Zenodo record 4562782) were read in R, and each count matrix (genes × cells) and its doublet annotation were written to Matrix Market and text files that both packages read. No cells or genes were removed; scDblFinder and pyscdblfinder received the same matrices.

Runs. In R: `set.seed(seed)` and `scDblFinder(sce)` on a `SingleCellExperiment` with the counts, with default arguments. In Python: `sc_dbl_finder(counts, random_state=seed)` with default arguments. Each package ran once per dataset with seed 1 and once with seed 2.

Evaluation. AUPRC (average precision) and AUROC were computed with scikit-learn from the doublet scores and the annotations. Agreement between Python and R is the mean over the four pairs of a Python seed and an R seed; agreement within R compares seed 1 with seed 2. Spearman correlations are over the scores of all cells of a dataset, and Jaccard indices over the sets of cells called doublets.

Embedding (Fig. 1f–i). Counts of pbmc-2ctrl-dm were normalised to 10,000 per cell and log-transformed with scanpy[6]; the 2,000 most variable genes were reduced to 30 principal components, and UMAP was computed on a 15-nearest-neighbour graph. The embedding is for display only; both packages scored the raw counts.

Run time. Run times are wall-clock times of the call above, without reading and writing files. Every run was pinned to its own processor core with BLAS and OpenMP limited to one thread; R and Python runs were spread over the same cores of one node at the same time, so that both packages were timed under the same load.

Notes for users coming from R

1. **Versions.** pyscdblfinder 0.3.x tracks scDblFinder 1.16.0, including its defaults (an expected doublet rate of 1% per thousand cells, 1,352 genes, 20 principal components, three classifier iterations).

2. **Matrix orientation.** `sc_dbl_finder` takes a features × cells matrix, as scDblFinder does; the `ScDblFinder` class takes an AnnData object (cells × genes) with raw counts in `X` or in a layer.

3. **Seeds.** `random_state` plays the role of `set.seed`. The same seed in R and Python gives different artificial doublets, so results are compared across seeds.

4. **Several samples.** pyscdblfinder runs one sample at a time; `ov.pp.qc(..., batch_key=...)` runs it per sample, as scDblFinder's `samples` argument does by default.